

2010
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

(a) Use the table of standard integrals to find $\int \frac{1}{\sqrt{4-x^2}} dx$. **1**

(b) Let $f(x) = \cos^{-1}\left(\frac{x}{2}\right)$. What is the domain of $f(x)$? **1**

(c) Solve $\ln(x+6) = 2 \ln x$. **3**

(d) Solve $\frac{3}{x+2} < 4$. **3**

(e) Use the substitution $u = 1 - x$ to evaluate $\int_0^1 x\sqrt{1-x} \, dx$. **3**

(f) Five ordinary six-sided dice are thrown. **1**

What is the probability that exactly two of the dice land showing a four?
Leave your answer in unsimplified form.

Question 2 (12 marks) Use a SEPARATE writing booklet.

- (a) The derivative of a function $f(x)$ is given by **2**

$$f'(x) = \sin^2 x.$$

Find $f(x)$, given that $f(0) = 2$.

- (b) The mass M of a whale is modelled by

$$M = 36 - 35.5e^{-kt},$$

where M is measured in tonnes, t is the age of the whale in years and k is a positive constant.

- (i) Show that the rate of growth of the mass of the whale is given by the differential equation **1**

$$\frac{dM}{dt} = k(36 - M).$$

- (ii) When the whale is 10 years old its mass is 20 tonnes. **2**

Find the value of k , correct to three decimal places.

- (iii) According to this model, what is the limiting mass of the whale? **1**

Question 2 continues on page 4

Question 2 (continued)

(c) Let

$$P(x) = (x + 1)(x - 3) Q(x) + ax + b,$$

where $Q(x)$ is a polynomial and a and b are real numbers.

The polynomial $P(x)$ has a factor of $x - 3$.

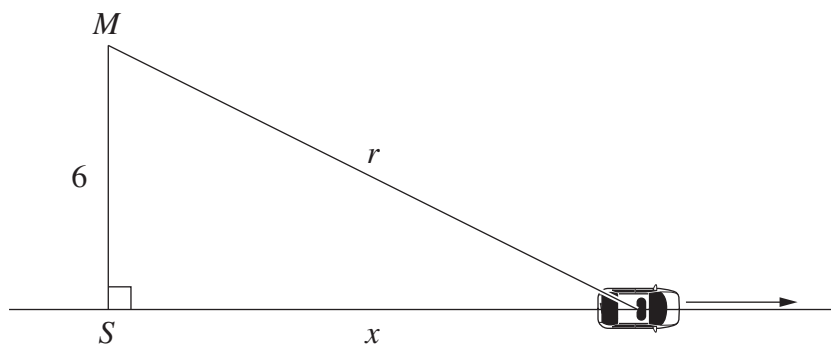
When $P(x)$ is divided by $x + 1$ the remainder is 8.

(i) Find the values of a and b . **2**

(ii) Find the remainder when $P(x)$ is divided by $(x + 1)(x - 3)$. **1**

(d) A radio transmitter M is situated 6 km from a straight road. The closest point on the road to the transmitter is S . **3**

A car is travelling away from S along the road at a speed of 100 km h^{-1} . The distance from the car to S is x km and from the car to M is r km.



Find an expression in terms of x for $\frac{dr}{dt}$, where t is time in hours.

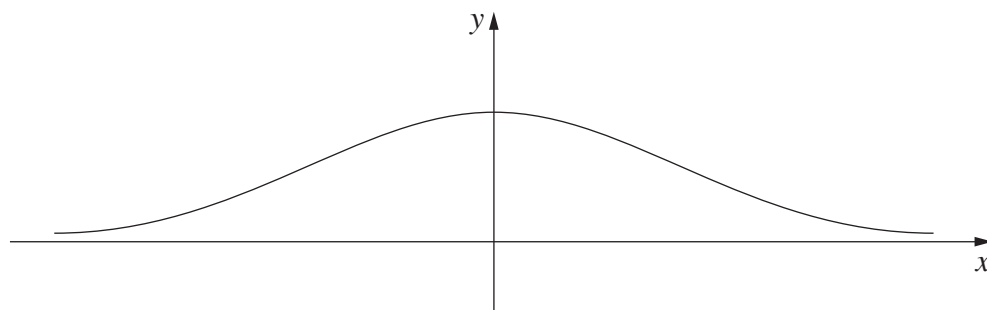
End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

- (a) At the front of a building there are five garage doors. Two of the doors are to be painted red, one is to be painted green, one blue and one orange.

- (i) How many possible arrangements are there for the colours on the doors? **1**
- (ii) How many possible arrangements are there for the colours on the doors if the two red doors are next to each other? **1**

- (b) Let $f(x) = e^{-x^2}$. The diagram shows the graph $y = f(x)$.



- (i) The graph has two points of inflexion. **3**
Find the x coordinates of these points.
- (ii) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. **1**
- (iii) Find a formula for $f^{-1}(x)$ if the domain of $f(x)$ is restricted to $x \geq 0$. **2**
- (iv) State the domain of $f^{-1}(x)$. **1**
- (v) Sketch the curve $y = f^{-1}(x)$. **1**
- (vi) (1) Show that there is a solution to the equation $x = e^{-x^2}$ between $x = 0.6$ and $x = 0.7$. **1**
- (2) By halving the interval, find the solution correct to one decimal place. **1**

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) A particle is moving in simple harmonic motion along the x -axis.

Its velocity v , at x , is given by $v^2 = 24 - 8x - 2x^2$.

- (i) Find all values of x for which the particle is at rest. **1**
- (ii) Find an expression for the acceleration of the particle, in terms of x . **1**
- (iii) Find the maximum speed of the particle. **2**

- (b) (i) Express $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right)$ in the form $R\cos(\theta + \alpha)$, **3**

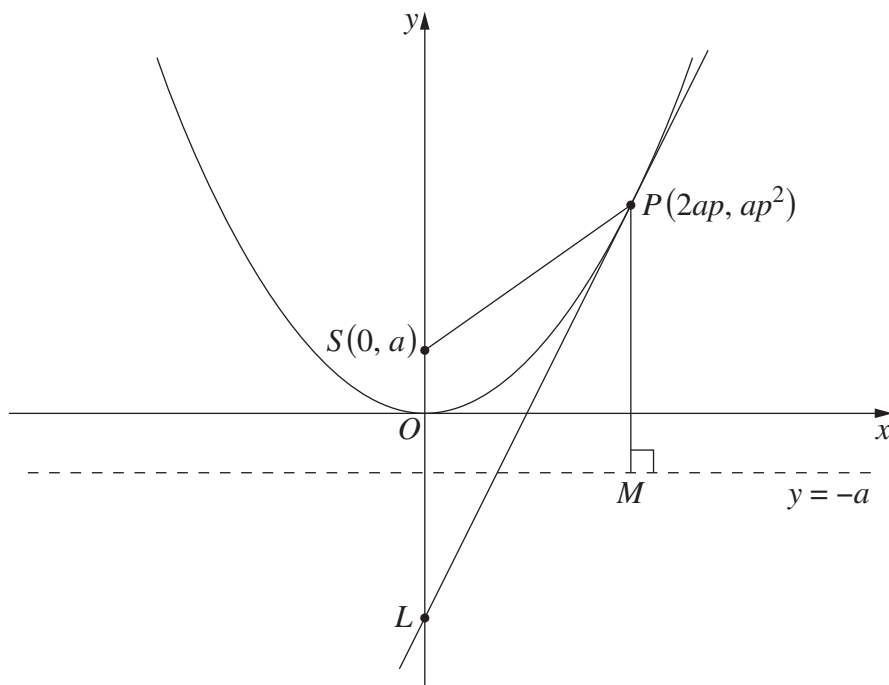
where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

- (ii) Hence, or otherwise, solve $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right) = 3$, **2**
for $0 < \theta < 2\pi$.

Question 4 continues on page 7

Question 4 (continued)

- (c) The diagram shows the parabola $x^2 = 4ay$. The point $P(2ap, ap^2)$, where $p \neq 0$, is on the parabola. 3



The tangent to the parabola at P , $y = px - ap^2$, meets the y -axis at L .

The point M is on the directrix, such that PM is perpendicular to the directrix.

Show that $SLMP$ is a rhombus.

End of Question 4

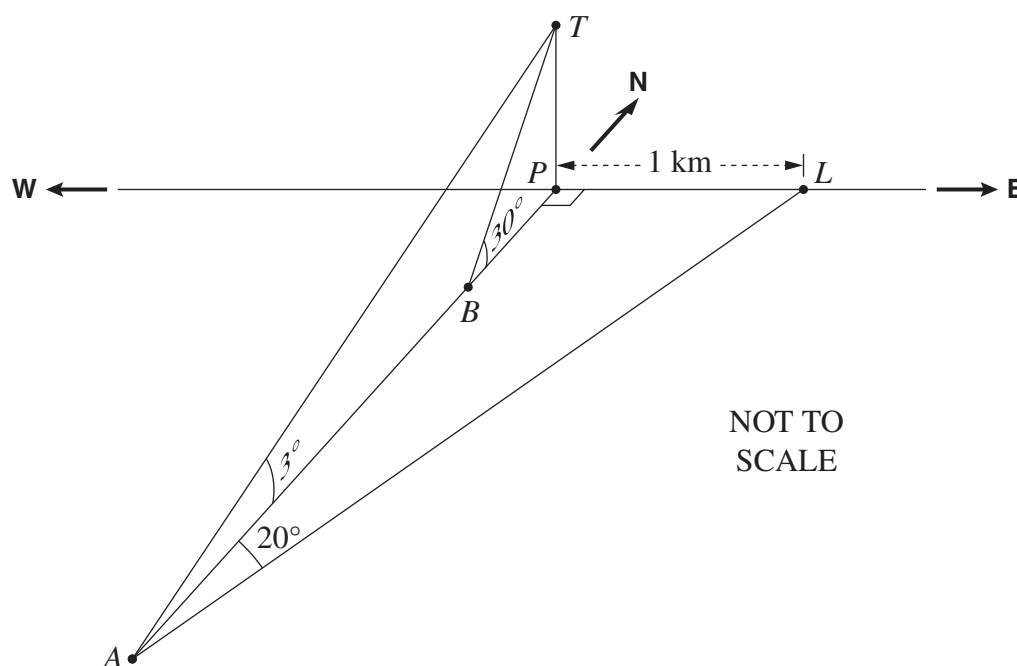
Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) A boat is sailing due north from a point A towards a point P on the shore line. The shore line runs from west to east.

In the diagram, T represents a tree on a cliff vertically above P , and L represents a landmark on the shore. The distance PL is 1 km.

From A the point L is on a bearing of 020° , and the angle of elevation to T is 3° .

After sailing for some time the boat reaches a point B , from which the angle of elevation to T is 30° .



(i) Show that $BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$. **3**

- (ii) Find the distance AB . **1**

Question 5 continues on page 9

Question 5 (continued)

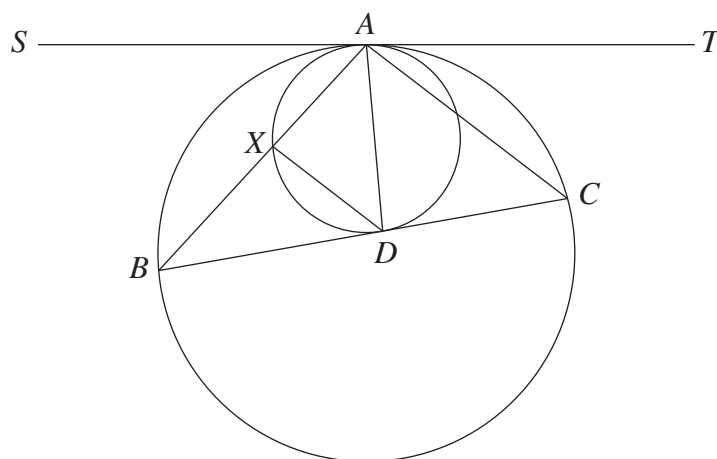
(b) Let $f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$ for $x \neq 0$.

(i) By differentiating $f(x)$, or otherwise, show that $f(x) = \frac{\pi}{2}$ for $x > 0$. **3**

(ii) Given that $f(x)$ is an odd function, sketch the graph $y = f(x)$. **1**

(c) In the diagram, ST is tangent to both the circles at A .

The points B and C are on the larger circle, and the line BC is tangent to the smaller circle at D . The line AB intersects the smaller circle at X .



Copy or trace the diagram into your writing booklet.

(i) Explain why $\angle AXD = \angle ABD + \angle XDB$. **1**

(ii) Explain why $\angle AXD = \angle TAC + \angle CAD$. **1**

(iii) Hence show that AD bisects $\angle BAC$. **2**

End of Question 5

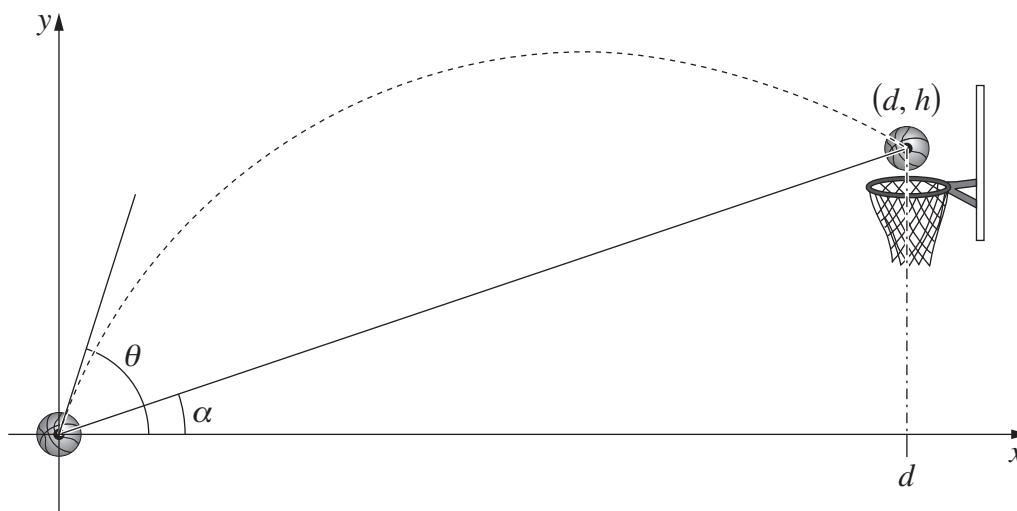
Question 6 (12 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $\cos(A - B) = \cos A \cos B(1 + \tan A \tan B)$. 1

(ii) Suppose that $0 < B < \frac{\pi}{2}$ and $B < A < \pi$. 1

Deduce that if $\tan A \tan B = -1$, then $A - B = \frac{\pi}{2}$.

- (b) A basketball player throws a ball with an initial velocity $v \text{ m s}^{-1}$ at an angle θ to the horizontal. At the time the ball is released its centre is at $(0, 0)$, and the player is aiming for the point (d, h) as shown on the diagram. The line joining $(0, 0)$ and (d, h) makes an angle α with the horizontal, where $0 < \alpha < \theta < \frac{\pi}{2}$.



Assume that at time t seconds after the ball is thrown its centre is at the point (x, y) , where

$$x = vt \cos \theta$$

$$y = vt \sin \theta - 5t^2.$$

(You are NOT required to prove these equations.)

Question 6 continues on page 11

Question 6 (continued)

- (i) If the centre of the ball passes through (d, h) show that **3**

$$v^2 = \frac{5d}{\cos\theta \sin\theta - \cos^2\theta \tan\alpha}.$$

- (ii) (1) What happens to v as $\theta \rightarrow \alpha$? **1**

- (2) What happens to v as $\theta \rightarrow \frac{\pi}{2}$? **1**

- (iii) For a fixed value of α , let $F(\theta) = \cos\theta \sin\theta - \cos^2\theta \tan\alpha$. **2**

Show that $F'(\theta) = 0$ when $\tan 2\theta \tan \alpha = -1$.

- (iv) Using part (a) (ii) or otherwise show that $F'(\theta) = 0$ when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$. **1**

- (v) Explain why v^2 is a minimum when $\theta = \frac{\alpha}{2} + \frac{\pi}{4}$. **2**

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

(a) Prove by induction that

3

$$47^n + 53 \times 147^{n-1}$$

is divisible by 100 for all integers $n \geq 1$.

(b) The binomial theorem states that

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n.$$

(i) Show that $2^n = \sum_{k=0}^n \binom{n}{k}$.

1

(ii) Hence, or otherwise, find the value of

1

$$\binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100}.$$

(iii) Show that $n2^{n-1} = \sum_{k=1}^n k \binom{n}{k}$.

2

Question 7 continues on page 13

Question 7 (continued)

- (c) (i) A box contains n identical red balls and n identical blue balls. A selection of r balls is made from the box, where $0 \leq r \leq n$. **1**

Explain why the number of possible colour combinations is $r + 1$.

- (ii) Another box contains n white balls labelled consecutively from 1 to n . **1**
A selection of $n - r$ balls is made from the box, where $0 \leq r \leq n$.

Explain why the number of different selections is $\binom{n}{r}$.

- (iii) The n red balls, the n blue balls and the n white labelled balls are all placed into one box, and a selection of n balls is made. **3**

Using part (b), or otherwise, show that the number of different selections is $(n + 2)2^{n-1}$.

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2010 HSC Mathematics Extension 1 Marking Guidelines

Question 1 (a)

Criteria	Marks
• Correct primitive	1

Question 1 (b)

Criteria	Marks
• Correct answer	1

Question 1 (c)

Criteria	Marks
• Correct solution	3
• Obtains $x = 3, -2$, or equivalent merit	2
• Attempts to obtain $x + 6 = x^2$, or equivalent merit	1

Question 1 (d)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Recognises an appropriate method	1

Question 1 (e)

Criteria	Marks
• Correct solution	3
• Correct primitive in terms of u , or equivalent merit	2
• Obtains $-(1-u)\sqrt{u}$ as the integrand, or equivalent merit	1

Question 1 (f)

Criteria	Marks
• Correct answer	1

Question 2 (a)

Criteria	Marks
• Correct solution	2
• Uses the identity $\sin^2 x = \frac{1 - \cos 2x}{2}$, or equivalent merit	1

Question 2 (b) (i)

Criteria	Marks
• Correct solution	1

Question 2 (b) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to solve $20 = 36 - 35.5e^{-10k}$	1

Question 2 (b) (iii)

Criteria	Marks
• Correct answer	1

Question 2 (c) (i)

Criteria	Marks
• Correctly finds a and b	2
• Shows an understanding of the remainder theorem	1

Question 2 (c) (ii)

Criteria	Marks
• Correct answer	1

Question 2 (d)

Criteria	Marks
• Correct solution	3
• Finds $\frac{dr}{dx}$ in terms of x , or equivalent merit	2
• Recognises $\frac{dr}{dt} = \frac{dr}{dx} \cdot \frac{dx}{dt}$ OR $r^2 = x^2 + 36$, or equivalent merit	1

Question 3 (a) (i)

Criteria	Marks
• Correct answer	1

Question 3 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 3 (b) (i)

Criteria	Marks
• Finds x coordinates correctly	3
• Finds $f''(x)$ correctly, or equivalent merit	2
• Finds $f'(x)$ correctly, or equivalent merit	1

Question 3 (b) (ii)

Criteria	Marks
• Correct explanation	1

Question 3 (b) (iii)

Criteria	Marks
• Finds $f^{-1}(x) = \sqrt{-\ln x}$	2
• Makes some progress by correctly taking logarithms of both sides	1

Question 3 (b) (iv)

Criteria	Marks
• Correct answer, ie $0 < x \leq 1$	1

Question 3 (b) (v)

Criteria	Marks
• Correct graph	1

Question 3 (b) (vi) (1)

Criteria	Marks
• Correct solution	1

Question 3 (b) (vi) (2)

Criteria	Marks
• Substitutes 0.65 and draws a correct conclusion	1

Question 4 (a) (i)

Criteria	Marks
• Correct answer	1

Question 4 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 4 (a) (iii)

Criteria	Marks
• Correct solution	2
• Finds that the maximum value occurs at $x = -2$, or equivalent merit	1

Question 4 (b) (i)

Criteria	Marks
• Correct answer	3
• Correctly finds either a value for R , or a value for α	2
• Expresses $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right)$ as $3\cos\theta - \sqrt{3}\sin\theta$, or equivalent merit	1

Question 4 (b) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to solve the equation using the result from part (i), or equivalent merit	1

Question 4 (c)

Criteria	Marks
• Correct proof	3
• Makes significant progress	2
• Finds one piece of relevant information	1

Question 5 (a) (i)

Criteria	Marks
• Correct solution	3
• Finds relevant expressions for TWO of AP , PT , BP	2
• Finds a relevant expression for ONE of AP , PT , BP	1

Question 5 (a) (ii)

Criteria	Marks
• Correct solution	1

Question 5 (b) (i)

Criteria	Marks
• Correct solution	3
• Correctly shows that $f'(x) = 0$, or equivalent merit	2
• Attempts to differentiate $f(x)$, or equivalent merit	1

Question 5 (b) (ii)

Criteria	Marks
• Correct graph	1

Question 5 (c) (i)

Criteria	Marks
• Correct explanation	1

Question 5 (c) (ii)

Criteria	Marks
• Correct explanation	1

Question 5 (c) (iii)

Criteria	Marks
• Correct proof	2
• Shows that $\angle TAC = \angle ABD$, or that $\angle XDB = \angle BAD$, or equivalent merit	1

Question 6 (a) (i)

Criteria	Marks
• Correct solution	1

Question 6 (a) (ii)

Criteria	Marks
• Correct deduction	1

Question 6 (b) (i)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Uses $(x, y) = (d, h)$ and attempts to eliminate t	1

Question 6 (b) (ii) (1)

Criteria	Marks
• Correct answer	1

Question 6 (b) (ii) (2)

Criteria	Marks
• Correct answer	1

Question 6 (b) (iii)

Criteria	Marks
• Correct solution	2
• Correctly differentiates $F(\theta)$	1

Question 6 (b) (iv)

Criteria	Marks
• Correct solution	1

Question 6 (b) (v)

Criteria	Marks
• Correct explanation	2
• Recognises that v^2 is a minimum when $F(\theta)$ is a maximum, or equivalent merit	1

Question 7 (a)

Criteria	Marks
• Correct proof	3
• Makes substantial progress	2
• Shows the statement is true for $n = 1$, or equivalent merit	1

Question 7 (b) (i)

Criteria	Marks
• Substitutes $x = 1$	1

Question 7 (b) (ii)

Criteria	Marks
• Correct answer	1

Question 7 (b) (iii)

Criteria	Marks
• Correct solution	2
• Differentiates both sides of $(1+x)^n = \binom{n}{0} + \dots + \binom{n}{n}x^n$ correctly, or equivalent merit	1

Question 7 (c) (i)

Criteria	Marks
• Correct explanation	1

Question 7 (c) (ii)

Criteria	Marks
• Correct explanation	1

Question 7 (c) (iii)

Criteria	Marks
• Correct solution	3
• Writes an expression summing from $r = 0$ to n , or equivalent progress	2
• Determines number of selections for a particular r , or equivalent merit	1

Mathematics Extension 1

2010 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	1	15.5E	HE4
1 (b)	1	15.2E	HE4
1 (c)	3	12.2	H3
1 (d)	3	1.4E	PE3
1 (e)	3	11.5E	HE4
1 (f)	1	18.2E	HE3
2 (a)	2	13.6E	HE6, H5
2 (b) (i)	1	14.2E	HE3
2 (b) (ii)	2	14.2E	HE3
2 (b) (iii)	1	14.2E	HE3, HE7
2 (c) (i)	2	16.2E	PE3
2 (c) (ii)	1	16.2E	PE3
2 (d)	3	14.1E	HE5
3 (a) (i)	1	18.1E	PE3
3 (a) (ii)	1	18.1E	PE3
3 (b) (i)	3	10.4, 12.5	H3, H5
3 (b) (ii)	1	15.1E	HE4, HE7
3 (b) (iii)	2	15.1E, 12.2, 12.3	HE4, H3
3 (b) (iv)	1	15.1E	HE4
3 (b) (v)	1	15.1E	HE4
3 (b) (vi) (1)	1	16.4E	HE7
3 (b) (vi) (2)	1	16.4E	HE7
4 (a) (i)	1	14.3E	HE3
4 (a) (ii)	1	14.3E	HE3
4 (a) (iii)	2	14.3E	HE3
4 (b) (i)	3	5.7E, 5.9E, 13.1	HE3
4 (b) (ii)	2	5.7E, 5.9E, 13.1	HE3, HE7, H5
4 (c)	3	9.6E, 6.8	PE3, PE4, HE7
5 (a) (i)	3	5.6E	PE2, PE6

Question	Marks	Content	Syllabus outcomes
5 (a) (ii)	1	5.6E	PE2, PE6
5 (b) (i)	3	15.5E, 15.4E	HE4, HE5
5 (b) (ii)	1	4.2, 15.2E, 15.3E	HE4, HE7
5 (c) (i)	1	2.10	PE2, PE3, HE7
5 (c) (ii)	1	2.10	PE2, PE3, HE7
5 (c) (iii)	2	2.10	PE2, PE3, HE7
6 (a) (i)	1	5.7E	PE2, HE7
6 (a) (ii)	1	5.7E, 5.9E, 13.1	PE2, HE7
6 (b) (i)	3	14.3E	HE3, HE7
6 (b) (ii) (1)	1	14.3E	HE3, HE7
6 (b) (ii) (2)	1	13.4, 14.3E	HE3, HE7
6 (b) (iii)	2	13.5	HE3, HE7
6 (b) (iv)	1	5.7E	HE3, HE7
6 (b) (v)	2	14.3E	HE3, HE7
7 (a)	3	7.4E	HE2
7 (b) (i)	1	17.3E	HE7
7 (b) (ii)	1	17.3E	HE7
7 (b) (iii)	2	17.3E	HE7
7 (c) (i)	1	18.1E	PE2, PE3
7 (c) (ii)	1	18.1E	PE2, PE3
7 (c) (iii)	3	18.1E	PE2, PE3